

	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^-}^{\#1} \alpha\beta_X$						
$\mathcal{K}_{0^+}^{\#1} \dagger$	0	0						
$\mathcal{K}_{0^+}^{\#1} \dagger^{\alpha\beta_X}$	0	0	$\mathcal{K}_{1^+}^{\#1} \alpha\beta$	$\mathcal{K}_{1^+}^{\#2} \alpha\beta$	$\mathcal{K}_{1^+}^{\#1} \alpha$	$\mathcal{K}_{1^+}^{\#2} \alpha$		
	$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta}$	- $\frac{\gamma_1 k^2}{2}$	$\frac{k^2 \binom{4}{K}_4}{2\sqrt{2}}$	0	0			
	$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$\frac{k^2 \binom{4}{K}_4}{2\sqrt{2}}$	$\frac{\gamma_2 k^2}{2}$	0	0			
	$\mathcal{K}_{1^+}^{\#1} \dagger^\alpha$	0	0	$-k^2 \binom{4}{K}_2$	$\frac{k^2 \binom{4}{K}_2}{\sqrt{2}}$			
	$\mathcal{K}_{1^+}^{\#2} \dagger^\alpha$	0	0	$\frac{k^2 \binom{4}{K}_2}{\sqrt{2}}$	$-\frac{k^2 \binom{4}{K}_2}{2}$	$\mathcal{K}_{2^+}^{\#1} \alpha\beta$	$\mathcal{K}_{2^+}^{\#1} \alpha\beta_X$	
					$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{3\gamma_3 k^2}{2}$	0	
					$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta_X}$	0	0	

	$\mathcal{T}_{0^+}^{\#1}$	$\mathcal{T}_{0^+}^{\#1}{}_{\alpha\beta\chi}$				
$\mathcal{T}_{0^+}^{\#1}{}^\dagger$	0	0				
$\mathcal{T}_{0^+}^{\#1}{}^\dagger{}_{\alpha\beta\chi}$	0	0	$\mathcal{T}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{T}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{T}_{1^+}^{\#1}{}_{\alpha}$	$\mathcal{T}_{1^+}^{\#2}{}_{\alpha}$
	$\mathcal{T}_{1^+}^{\#1}{}^\dagger{}_{\alpha\beta}$	$\frac{\Upsilon_2 k^2}{2 \text{Det}(1^+)}$	$-\frac{k^2 \binom{4}{\kappa_4}}{2 \sqrt{2} \text{Det}(1^+)}$	0	0	
	$\mathcal{T}_{1^+}^{\#2}{}^\dagger{}_{\alpha\beta}$	$-\frac{k^2 \binom{4}{\kappa_4}}{2 \sqrt{2} \text{Det}(1^+)}$	$-\frac{\Upsilon_1 k^2}{2 \text{Det}(1^+)}$	0	0	
	$\mathcal{T}_{1^+}^{\#1}{}^\dagger{}_{\alpha}$	0	0	$-\frac{4}{9 k^2 \binom{4}{\kappa_2}}$	$\frac{2 \sqrt{2}}{9 k^2 \binom{4}{\kappa_2}}$	
	$\mathcal{T}_{1^+}^{\#2}{}^\dagger{}_{\alpha}$	0	0	$\frac{2 \sqrt{2}}{9 k^2 \binom{4}{\kappa_2}}$	$-\frac{2}{9 k^2 \binom{4}{\kappa_2}}$	
					$\mathcal{T}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{T}_{2^+}^{\#1}{}_{\alpha\beta\chi}$
				$\mathcal{T}_{2^+}^{\#1}{}^\dagger{}_{\alpha\beta}$	$\frac{1}{\text{Det}(2^+)}$	0
				$\mathcal{T}_{2^+}^{\#1}{}^\dagger{}_{\alpha\beta\chi}$	0	0

Abbreviations used in matrices	
$Y_1 = 3 \left(\overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2} \right) + 2 \overset{(4)}{\kappa_3}$	$\&\& Y_2 = \overset{(4)}{\kappa_3} + \overset{(4)}{\kappa_4}$
$\&\& Y_3 = \overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2}$	$\&\&$
$\text{Det}(1^+) = -\frac{1}{8} k^4 \left(6 \overset{(4)}{\kappa_1} \left(\overset{(4)}{\kappa_3} + \overset{(4)}{\kappa_4} \right) + 6 \overset{(4)}{\kappa_2} \left(\overset{(4)}{\kappa_3} + \overset{(4)}{\kappa_4} \right) + \left(2 \overset{(4)}{\kappa_3} + \overset{(4)}{\kappa_4} \right)^2 \right)$	
$\&\& \text{Det}(2^+) = \frac{3}{2} k^2 \left(\overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2} \right)$	

$$\begin{aligned} & \text{Lagrangian} \\ & \kappa_1^{(4)} \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha\beta} + \kappa_2^{(4)} \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha\beta} + \kappa_3^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\alpha\chi} + \kappa_4^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} - 3 \kappa_1^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\chi\alpha} - \\ & 3 \kappa_2^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\chi\alpha} - \kappa_3^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\chi\alpha} + \frac{1}{2} \kappa_3^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \frac{1}{2} \kappa_4^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} \end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{T}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{T}_0^{\#1\alpha\beta\chi} == 0$	1	$\partial_\delta \partial^\alpha \mathcal{T}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{T}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{T}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{T}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{T}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{T}^{\beta\alpha\chi} ==$ $\partial_\delta \partial^\alpha \mathcal{T}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{T}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{T}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{T}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{T}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{T}^{\chi\alpha\beta}$
$\mathcal{T}_0^{\#1} == 0$	1	$\partial_\beta \mathcal{T}^\alpha{}_\alpha{}^\beta == 0$
$\mathcal{T}_1^{\#1\alpha} + 2 \mathcal{T}_1^{\#2\alpha} == 0$	3	$\partial_\chi \partial^\alpha \mathcal{T}^\beta{}_\beta{}^\chi + \partial_\chi \partial^\chi \mathcal{T}^{\beta\alpha}{}_\beta == 3 \partial_\chi \partial_\beta \mathcal{T}^{\beta\alpha\chi}$
$\mathcal{T}_2^{\#1\alpha\beta\chi} == 0$	5	$3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{T}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{T}^{\delta\beta}{}_\delta + 2 \partial_\epsilon \partial^\delta \partial_\delta \partial^\beta \mathcal{T}^{\alpha\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{T}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{T}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{T}^{\beta\alpha\delta} +$ $4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{T}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{T}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{T}^{\delta}{}_\delta{}^\epsilon + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{T}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{T}^{\delta\alpha}{}_\delta ==$ $3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{T}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{T}^{\delta\alpha}{}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{T}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{T}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{T}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{T}^{\alpha\beta\delta} +$ $2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{T}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{T}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{T}^{\delta}{}_\delta{}^\epsilon + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{T}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{T}^{\delta\beta}{}_\delta$

Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$ \begin{aligned} & (54 \kappa_2^{(4)3} + 54 \kappa_2^{(4)2} \kappa_3^{(4)} + 20 \kappa_2^{(4)} \kappa_3^{(4)2} + 36 \kappa_2^{(4)2} \kappa_4^{(4)} + 20 \kappa_2^{(4)} \kappa_3^{(4)} \kappa_4^{(4)} + 5 \kappa_2^{(4)} \kappa_4^{(4)2} + \\ & 18 \kappa_1^{(4)2} (3 \kappa_2^{(4)} + \kappa_3^{(4)} + \kappa_4^{(4)}) + 3 \kappa_1^{(4)} (36 \kappa_2^{(4)2} + (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 + 6 \kappa_2^{(4)} (4 \kappa_3^{(4)} + 3 \kappa_4^{(4)})) - \\ & \sqrt{(324 \kappa_1^{(4)4} (9 \kappa_2^{(4)2} - 2 \kappa_2^{(4)} (\kappa_3^{(4)} + \kappa_4^{(4)}) + (\kappa_3^{(4)} + \kappa_4^{(4)})^2) + \\ & \kappa_2^{(4)2} (2916 \kappa_2^{(4)4} + 7 (2 \kappa_3^{(4)} + \kappa_4^{(4)})^4 + 648 \kappa_2^{(4)3} (5 \kappa_3^{(4)} + 2 \kappa_4^{(4)}) + 36 \kappa_2^{(4)} (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 (4 \kappa_3^{(4)} + 3 \kappa_4^{(4)}) + \\ & 108 \kappa_2^{(4)2} (15 \kappa_3^{(4)2} + 16 \kappa_3^{(4)} \kappa_4^{(4)} + 5 \kappa_4^{(4)2})) + 9 \kappa_1^{(4)2} (1944 \kappa_2^{(4)4} + (2 \kappa_3^{(4)} + \kappa_4^{(4)})^4 + \\ & 216 \kappa_2^{(4)3} (4 \kappa_3^{(4)} + \kappa_4^{(4)}) + 4 \kappa_2^{(4)} (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 (6 \kappa_3^{(4)} + 7 \kappa_4^{(4)}) + 24 \kappa_2^{(4)2} (7 \kappa_3^{(4)2} + 13 \kappa_3^{(4)} \kappa_4^{(4)} + 7 \kappa_4^{(4)2})) + \\ & 108 \kappa_1^{(4)3} (108 \kappa_2^{(4)3} + 6 \kappa_2^{(4)2} (2 \kappa_3^{(4)} - \kappa_4^{(4)}) + (\kappa_3^{(4)} + \kappa_4^{(4)}) (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 + \kappa_2^{(4)} (4 \kappa_3^{(4)2} + 14 \kappa_3^{(4)} \kappa_4^{(4)} + 9 \kappa_4^{(4)2})) + \\ & 12 \kappa_1^{(4)} \kappa_2^{(4)} (972 \kappa_2^{(4)4} + 21 \kappa_2^{(4)} (\kappa_3^{(4)} + \kappa_4^{(4)}) (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 + (2 \kappa_3^{(4)} + \kappa_4^{(4)})^4 + \\ & 54 \kappa_2^{(4)3} (14 \kappa_3^{(4)} + 5 \kappa_4^{(4)}) + 9 \kappa_2^{(4)2} (28 \kappa_3^{(4)2} + 34 \kappa_3^{(4)} \kappa_4^{(4)} + 13 \kappa_4^{(4)2}))) / \\ & (\kappa_2^{(4)} (\kappa_1^{(4)} + \kappa_2^{(4)}) (6 \kappa_1^{(4)} (\kappa_3^{(4)} + \kappa_4^{(4)}) + 6 \kappa_2^{(4)} (\kappa_3^{(4)} + \kappa_4^{(4)}) + (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2)) \end{aligned} $
	2	0	$ \begin{aligned} & (54 \kappa_2^{(4)3} + 54 \kappa_2^{(4)2} \kappa_3^{(4)} + 20 \kappa_2^{(4)} \kappa_3^{(4)2} + 36 \kappa_2^{(4)2} \kappa_4^{(4)} + 20 \kappa_2^{(4)} \kappa_3^{(4)} \kappa_4^{(4)} + 5 \kappa_2^{(4)} \kappa_4^{(4)2} + \\ & 18 \kappa_1^{(4)2} (3 \kappa_2^{(4)} + \kappa_3^{(4)} + \kappa_4^{(4)}) + 3 \kappa_1^{(4)} (36 \kappa_2^{(4)2} + (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 + 6 \kappa_2^{(4)} (4 \kappa_3^{(4)} + 3 \kappa_4^{(4)})) + \\ & \sqrt{(324 \kappa_1^{(4)4} (9 \kappa_2^{(4)2} - 2 \kappa_2^{(4)} (\kappa_3^{(4)} + \kappa_4^{(4)}) + (\kappa_3^{(4)} + \kappa_4^{(4)})^2) + \\ & \kappa_2^{(4)2} (2916 \kappa_2^{(4)4} + 7 (2 \kappa_3^{(4)} + \kappa_4^{(4)})^4 + 648 \kappa_2^{(4)3} (5 \kappa_3^{(4)} + 2 \kappa_4^{(4)}) + 36 \kappa_2^{(4)} (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 (4 \kappa_3^{(4)} + 3 \kappa_4^{(4)}) + \\ & 108 \kappa_2^{(4)2} (15 \kappa_3^{(4)2} + 16 \kappa_3^{(4)} \kappa_4^{(4)} + 5 \kappa_4^{(4)2})) + 9 \kappa_1^{(4)2} (1944 \kappa_2^{(4)4} + (2 \kappa_3^{(4)} + \kappa_4^{(4)})^4 + \\ & 216 \kappa_2^{(4)3} (4 \kappa_3^{(4)} + \kappa_4^{(4)}) + 4 \kappa_2^{(4)} (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 (6 \kappa_3^{(4)} + 7 \kappa_4^{(4)}) + 24 \kappa_2^{(4)2} (7 \kappa_3^{(4)2} + 13 \kappa_3^{(4)} \kappa_4^{(4)} + 7 \kappa_4^{(4)2})) + \\ & 108 \kappa_1^{(4)3} (108 \kappa_2^{(4)3} + 6 \kappa_2^{(4)2} (2 \kappa_3^{(4)} - \kappa_4^{(4)}) + (\kappa_3^{(4)} + \kappa_4^{(4)}) (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 + \kappa_2^{(4)} (4 \kappa_3^{(4)2} + 14 \kappa_3^{(4)} \kappa_4^{(4)} + 9 \kappa_4^{(4)2})) + \\ & 12 \kappa_1^{(4)} \kappa_2^{(4)} (972 \kappa_2^{(4)4} + 21 \kappa_2^{(4)} (\kappa_3^{(4)} + \kappa_4^{(4)}) (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 + (2 \kappa_3^{(4)} + \kappa_4^{(4)})^4 + \\ & 54 \kappa_2^{(4)3} (14 \kappa_3^{(4)} + 5 \kappa_4^{(4)}) + 9 \kappa_2^{(4)2} (28 \kappa_3^{(4)2} + 34 \kappa_3^{(4)} \kappa_4^{(4)} + 13 \kappa_4^{(4)2}))) / \\ & (\kappa_2^{(4)} (\kappa_1^{(4)} + \kappa_2^{(4)}) (6 \kappa_1^{(4)} (\kappa_3^{(4)} + \kappa_4^{(4)}) + 6 \kappa_2^{(4)} (\kappa_3^{(4)} + \kappa_4^{(4)}) + (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2)) \end{aligned} $

[illegible]